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ORIGINAL ARTICLE

ZOO ANIMALS

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Characterization of antimicrobial resistance profiles in *Escherichia coli* isolated from captive mammals in Ecuador

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Abstract

Background: This study focuses on the AMR profiles in E. coli isolated from captive mammals at EcoZoo San Martín, Baños de Agua Santa, Ecuador, highlighting the role of wildlife as reservoirs of resistant bacteria.

Aims: The aim of this research is to investigate the antimicrobial resistance profiles of E. coli strains isolated from various species of captive mammals, emphasizing the potential zoonotic risks and the necessity for integrated AMR management strategies.

Materials & Methods: A total of 189 fecal samples were collected from 70 mammals across 27 species. These samples were screened for E. coli, resulting in 90 identified strains. The resistance profiles of these strains to 16 antibiotics, including 10 β-lactams and 6 non-β-lactams, were determined using the disk diffusion method. Additionally, the presence of Extended-Spectrum Beta-Lactamase (ESBL) genes and other resistance genes was analyzed using PCR.

Results: Significant resistance was observed, with 52.22% of isolates resistant to ampicillin, 42.22% to ceftriaxone and cefuroxime, and 27.78% identified as ESBL-producing E. coli. Multiresistance (resistance to more than three antibiotic groups) was found in 35.56% of isolates. Carnivorous and omnivorous animals, particularly those with prior antibiotic treatments, were more likely to harbor resistant strains.

Discussion: These findings underscore the role of captive mammals as indicators of environmental AMR. The high prevalence of resistant E. coli in these animals suggests that zoos could be significant reservoirs for the spread of antibiotic-resistant bacteria. The results align with other studies showing that diet and antibiotic treatment history influence resistance profiles.

Conclusion: The study highlights the need for an integrated approach involving veterinary care, habitat management, and public awareness to prevent captive wildlife from becoming reservoirs of antibiotic-resistant bacteria. Improved waste management practices and responsible antibiotic use are crucial to mitigate the risks of AMR in zoo environments and reduce zoonotic threats.

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KEYWORDS
antimicrobial resistance, captive mammals, ESBL-producing bacteria, *Escherichia coli*, One Health, zoo management

1 | INTRODUCTION

In the 21st century, one of the most alarming threats to global public health is the pandemic of antimicrobial resistance (AMR) (United Nations Environment Programme, 2023). This crisis, often referred to as a 'silent pandemic', poses a significant challenge to health care systems worldwide (Mendelson et al., 2022). Unlike the immediate and visible impacts of infectious disease pandemics such as COVID-19, the AMR pandemic is insidious yet pervasive, undermining the efficacy of antibiotics and other antimicrobial agents that have been cornerstones of modern medicine (Walesch et al., 2023).

The rise in AMR can be attributed to several factors, including the overuse and misuse of antibiotics in human medicine, agriculture, and livestock (Irfan et al., 2022). These practices accelerate the evolution and spread of resistant pathogens, a phenomenon that has been exacerbated by the global interconnectedness of human, animal and environmental health (Hernando-Amado et al., 2019). The concept of One Health, which recognizes the interconnectedness of these health domains, is crucial in understanding and combating the spread of AMR (Calero-Cáceres et al., 2022).

Wildlife species are an integral part of the environmental compartment and play a significant role in the ecology of AMR (Dolejska, 2020). These species have the potential to naturally harbour antibiotic-resistant bacteria, thereby acting as reservoirs and vectors for the spread of resistance genes (Laborda et al., 2022). This aspect is particularly pertinent as wildlife can interact with different ecosystems, potentially facilitating the transfer of resistant bacteria across various environmental niches (Lee et al., 2022). The significance of wildlife in the context of AMR is further highlighted by the scarcity of focused research in this area. Studies on antibiotic resistance in wildlife and its environmental implications are limited, leading to a substantial gap in our understanding of the dynamics of AMR in natural ecosystems (Ramey & Ahlstrom, 2020).

Considering the critical importance of addressing AMR, this research aims to investigate AMR profiles in *Escherichia coli* isolated from captive mammals at the Eco Zoo 'San Martín' in Baños de Agua Santa. This study could serve as a model for understanding AMR in wildlife species, providing valuable insights into the broader context of the AMR pandemic. By focusing on a specific ecological niche, this research contributes to the broader effort to understand and mitigate the global challenge of AMR.

2 | MATERIALS AND METHODS

2.1 | Source of data and ethical approval

This study was conducted with the authorization of the Ministry of Environment of Ecuador (contract MAATE-DBI-CM-2023-0278). The study did not directly involve the use of animals, but rather their faeces, and the sampling process was assessed by zoo specialists to ensure minimal disturbance to the animals. Data were collected once per zone from captive mammals at San Martin Eco Zoo from April 2022 to June 2022. The variables extracted from the dataset included the common name, scientific name, order, species per zoo habitat, food habits, condition, time spent at the zoo and information regarding any antimicrobial treatment received.

2.2 | San martin eco zoo

San Martin Eco Zoo in Ecuador is a wildlife management centre, authorized by the Provincial Directorate of the Tungurahua Ministry of the Environment. The zoo serves as a refuge for wildlife affected by illegal species trafficking and is home to approximately 300 animals across 120 different species, including 23 mammals, 35 avian and 4 reptilian species. Spanning 8 hectares of tropical forest, the zoo is located 2 km from Baños de Agua Santa, Tungurahua province, and welcomes around 28,000 visitors annually.

2.3 | Sample collection

E. coli isolates were collected from the faecal samples of captive mammals in three different areas of the zoo. Sterile wood sticks and coprotainers were used for sample collection. Each coprotainer was labelled with information about the mammal species from the corresponding habitat. The samples were transported to the Research Laboratory at the Technical University of Ambato from the San Martin Eco Zoo between April 2022 and June 2022 at 4°C.

2.4 | Selective isolation and identification of *E. coli*

All samples were processed in the laboratory within 6 h of collection. Samples were homogenized in 10 mL of 1× PBS for 1 min.

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Subsequently, they were streaked on Eosin Methylene Blue agar (Oxoid). Additionally, CHROMagar mSuperCARBA (CHROMagar Microbiology) was used to detect carbapenem-resistant *E. coli*. Cultures were incubated at 37°C for 24 h. Suspect *E. coli* colonies from each Petri dish were purified and identified using Gram-staining and the following biochemical tests: a catalase test using 30% hydrogen peroxide (Merck Millipore); triple sugar iron agar test (Becton Dickinson GmbH); simmons citrate agar test (Merck); lysine iron agar (Merck); christensen urea agar test (Britania Lab.); indole reaction using tryptone water (Merck) and Kovac's reagent (Sigma-Aldrich). For each positive sample, one isolate was selected, cryopreserved in Luria Bertani (LB) broth (Sigma-Aldrich) supplemented with 30% glycerol (Merck Millipore), and maintained at −80°C until analysis.

2.5 | Antimicrobial susceptibility testing

Susceptibility to 16 antimicrobial agents was determined using the disk diffusion method on Muller-Hinton agar (Merck). AMC30: amoxicillin–clavulanic acid (30 μg), TZP100: piperacillin–tazobactam (110 μg), AM10: ampicillin (10 μg), FEP30: cefepime (30 μg), FOX30: cefoxitin (30 μg), CAZ30: ceftazidime (30 μg), CRO30: ceftriaxone (30 μg), CXM30: cefuroxime (30 μg), ETP10: ertapenem (10 μg), MEM10: meropenem (10 μg), AK30: amikacin (30 μg), CN10: gentamicin (10 μg), F300: nitrofurantoin (300 μg), FF200: fosfomycin (200 μg), SXT 1.25: trimethoprim–sulfamethoxazole (25 μg) and CIP5: ciprofloxacin (5 μg) (Bioanalyzer). Interpretations followed the Clinical and Laboratory Standards Institute standards (Clinical and Laboratory Standards Institute, 2023). *E. coli* isolates were tested for extended-spectrum beta-lactamase (ESBL) production using the double-disk synergy test (Dobiasova et al., 2013).

2.6 | Detection of antibiotic resistance genes

Polymerase chain reactions (PCR) were performed to identify genes associated with the ESBL phenotype and various antibiotic resistance categories. These categories included ESBL-producing genes (bla_{TEM}, bla_{SHV} and $bla_{CTX\text{-}M}$), carbapenemase-encoding genes ($bla_{OXA\text{-}48}$, bla_{NDM}, bla_{VIM}, bla_{KPC} and bla_{IMP}), genes related to mobile colistin resistance (*mcr-1*), and quinolone resistance genes (*qnrA* and *qnrS*). The specific primers and conditions for each gene are detailed in Table S1. The PCR products were analysed via electrophoresis in a 1.2% agarose gel stained with DNA gel stain (Thermo Fisher Scientific) and run at 90 V for 2 h using a 100 bp molecular weight marker to assess the presence of these resistance genes.

2.7 | Fingerprinting (GTG)-5

(GTG)-5-based genotyping was performed using a polytrinucleotide (GTG) primer targeting conserved poly GTG sequences in bacterial genomes (De Vuyst et al., 2008). For the execution of this technique, we used 12.5 μL of 2× Phusion U Green Multiplex PCR Master Mix (Thermo Scientific) mixed with 0.25 μL of primer (GTG)-5 (5′-GTG-GTG-GTG-GTG-GTG-3′) at a concentration of 30 μM, 11 μL of nuclease-free water and 1.25 μL of bacterial DNA from heat shock (Mohapatra et al., 2007). To obtain bacterial DNA, we performed a thermal shock (Albán et al., 2020). The reaction conditions were as follows: initial denaturation at 94°C for 5 min, 30 cycles of denaturation at 94°C for 30 s, annealing at 45°C for 1 min, elongation at 65°C for 8 min and a final elongation at 65°C for 16 min, followed by storage at 4°C until analysis. Electrophoresis was performed at 2% agarose (Invitrogen) using 1× TBE (Invitrogen) at 90 V for 5 h and a 1 kb molecular weight marker, then the gel was stained with DNA gel stain (Thermo Fisher Scientific). Gel images were analysed using GelJ software (Heras et al., 2015). The similarity between strains was determined using the Dice index, employing the unweighted pair group method with arithmetic mean algorithm. The generated Newick file was subsequently analysed using iTOL software to assess phylogenetic relationships (Letunic & Bork, 2021).

2.8 | Transconjugation assays

The transferability of resistance genes was tested through conjugation. To this end, experiments were conducted using sodium azide-resistant *E. coli* J53 (AziR) as the recipient and ESBL-positive *E. coli* as donors. *E. coli* J53 AziR were cultured on MacConkey agar supplemented with sodium azide (100 μg/mL) and donor *E. coli* strains on MacConkey agar supplemented with ceftriaxone (3 μg/mL) and incubating both at 37°C for 24 h (Calero-Cáceres et al., 2022). Subsequently, both strains were grown in LB Broth (Merck) supplemented with the respective concentrations of sodium azide and ceftriaxone for 24 h. When the cultures of *E. coli* J53 AziR and ESBL-positive *E. coli* reached the same optical density, equal volumes (0.5 mL) of the cultures were centrifuged at 5000 rpm for 5 min to wash the cells and remove the LB broth containing sodium azide and ceftriaxone. The cells were then resuspended in LB broth without supplements, and 0.5 mL of each strain was mixed in 4 mL of LB broth, followed by incubation at 37°C for 24 h. Transconjugants were subsequently selected using MacConkey plates containing sodium azide (100 μg/mL) and ceftriaxone (3 μg/mL). To evaluate the transfer of the ESBL plasmid, antimicrobial susceptibility testing and PCR were applied.

2.9 | Sequencing

Sanger sequencing was performed on PCR products of approximately 500 bp from the CTX-M gene using an ABI 3500xL Genetic Analyzer (Applied Biosystems) and a BigDye Terminator v3.1 Cycle Sequencing Kit. The CTX-M sequences were analysed using MEGA X Molecular Genetic and Evolutionary Analysis Version 10.2.6 and BioEdit Sequence Alignment Editor Version 7.2.5. The consensus CTX-M sequences were compared with the CARD antibiotic resistance database using the nt-BLAST tool. The CTX-M family

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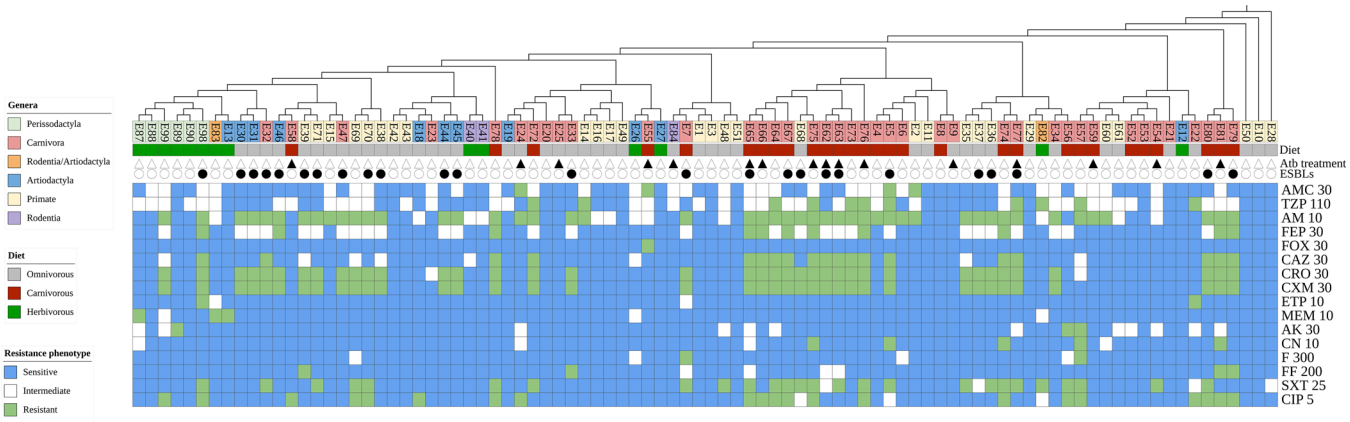


FIGURE 1 Dendrogram obtained by cluster analysis of fingerprint rep-PCR (GTG5) and complementary metadata. The dendrogram is based on the dice similarity coefficient with the unweighted pair group method using arithmetic averages (UPGMA) algorithm. PCR, Polymerase chain reaction.

classification was determined using the beta-lactamase database (Naas et al., 2017).

3 | RESULTS AND DISCUSSION

Out of 189 isolations, 90 were identified as *E. coli*. These strains were collected from 70 mammals, representing 27 different species. Table S2 provides details on the origin, habitat, common name and scientific name of the animals from which the isolates were obtained. For the GTG5 fingerprint dendrogram, we incorporated data regarding the diet and antibiotic treatment history of the host species from which the isolates were obtained. GTG5 profile shows a high genotypic diversity among the isolates. There was no correlation between clonal diversity, resistance phenotypic profiles or the species from which the isolates were obtained; however, several strains with the same phenotypic profile showed identical fingerprints (Figure 1).

Resistance profile to the 16 antibiotics evaluated (10 β-lactams and 6 non-β-lactams) is shown in Figure 1. Among 90 strains, 52.22% strains showed resistant to ampicillin, 42.22% to ceftriaxone and cefuroxime, 28.89% to trimethoprim/sulfamethoxazole, 23.33% to ciprofloxacin, 21.11% to ceftazidime, 12.22% to piperacillin/tazobactam and cefepime, 5.56% to gentamicin, 4.44% to amoxicillin/clavulanic acid and fosfomycin, 3.33% to amikacin and meropenem, 2.22% to nitrofurantoin and ertapenem and 1.11% to cefoxitin.

Among them, 27.78% (25/90) were ESBL-producing *E. coli*, and 35.56% (32/90) *E. coli* isolates showed multiresistance phenotype (resistant to more than three antibiotics groups). Notably, 9 of these ESBL-positive isolates originated from carnivorous animals, 15 from animals with a mixed diet and 1 from an herbivorous diet. Furthermore, three isolates displaying ESBL positivity were sourced from ocelots (*Leopardus pardalis*) that had undergone antibiotic treatment. This information is detailed in the dendrogram generated through cluster analysis of rep-PCR (GTG5) fingerprints. These results are consistent with similar studies showing that carnivorous and omnivorous species are more likely to harbour antibiotic-resistant bacteria compared to herbivorous species (Bamunusinghage et al., 2022; Milanović et al., 2019; Vittecoq et al., 2016).

The increased likelihood of carnivorous species harbouring antibiotic-resistant bacteria can be attributed to several factors (Treiber & Beranek-Knauer, 2021). First, the diet of these animals often includes other animals that may have been exposed to antibiotics, either through veterinary treatments or as a result of consuming antibiotic-laden feed. In Ecuador, the use of antibiotics in the production of broiler chickens and beef cattle is a practice that has promoted the spread of multidrug-resistant bacteria producing ESBL (Calero-Cáceres et al., 2020; Ortega-Paredes et al., 2020). In the evaluated zoo, the presence of raw animal products intended for the consumption of the zoo's mammals could represent a route of entry for resistant microorganisms into the animals' intestinal microbiota. This exposure can lead to the development and accumulation of resistant bacteria within the food chain (Economou & Gousia, 2015).

It is also important to consider the presence of ESBL producing *E. coli* in isolates from omnivorous and herbivorous mammals. Regarding vegetables and fruits intended for consumption in the zoo, these are purchased at the Wholesale Market in the city of Riobamba. A study conducted in Riobamba markets showed that *E. coli* strains isolated from fresh vegetables exhibit multidrug resistance (3–12 antibiotics); among the analysed vegetables, white cabbage, lettuce, carrots, radishes, celery, cucumber and beef tomatoes stand out, which are used in the diets of the zoo's mammals (Barragán-Fonseca et al., 2022). In Ecuador, studies focused on irrigation waters and their role in the presence of AMR in vegetables and fruits from 17 different provinces showed that, in vegetables like lettuce, cabbage, spinach, beef tomatoes, strawberries, melons, apples, bananas and watermelons, there was the presence of ESBL-producing *E. coli*, and these strains were resistant to ampicillin and cefazolin, cefotaxime, tetracycline and cefepime (Montero et al., 2021).

Among the 70 different mammals' faeces analysed, specific cases included a squirrel (*Sciurus vulgaris*) and a kinkajou (*Potos flavus*), which had received antibiotic treatments within a 12-month period at the zoo. The squirrel, housed in the quarantine area, was treated with

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enrofloxacin for 7 days, whereas the kinkajou received a combination of enrofloxacin, meloxicam and doxycycline for 5 days before being transferred to habitat 10 in the mammal's section. Considering the distribution of captive mammals around the zoo and that different species share space and environment, isolations origin was assembled considering the order; however, in one habitat, species of two different orders lived together.

The molecular studies revealed that 7 isolates carried bla_{TEM} (4 *E. coli* from *Panthera tigris tigris*, 2 from *Panthera onca* and 1 from *P. flavus*), and 25 isolates were positive for $bla_{CTX\text{-}M}$ (10 *E. coli* from *L. pardalis*; 5 from *Cebus albifrons aequatorialis*; 2 from *Cebus capucinus*, *Callicebus discolor*, *P. tigris tigris* and *Tapirus terrestris*; 1 from *P. onca*, *Eira barbara*, *Dicotyles tajacu* and *Procyon cancrivorus*). No strains were positive for β-lactamases (bla_{SHV}, bla_{CMY}), carbapenemases, quinolones and colistin resistance genes. Positive CTX-M strains were screening by gene-specific primers and were CTX-M-1 group positive (16/25) and CTX-M-9 group positive (9/25) (Table 1).

Our findings reveal variable resistance profiles across different species, marking this as the higher report on antibiotic resistance in healthy captive mammalian *E. coli* isolates within Ecuadorian zoos. A notable comparison can be drawn to a parallel study in the Napo Province, which identified significant resistance to ampicillin and trimethoprim/sulfamethoxazole in Humboldt's white-fronted capuchins (*Cebus yuracus*) (Haro-León et al., 2020). However, that study was only focused on one animal species.

Previous studies in countries such as Peru and Colombia have demonstrated that captive species such as *Ateles belzebuth*, *Ateles chamek*, *Callicebus oenanthe*, *Lagothrix cana* and *Lagothrix lagotricha* host strains of *E. coli* with high resistance to cephalothin (KF) (46.2%), ampicillin/clavulanic acid (31.1%), tobramycin (TOB) (30.2%) and tetracycline (24.5%), and with high sensitivity to ceftriaxone (93.3%), ampicillin/sulbactam (SAM) (86.7%), trimethoprim/sulfamethoxazole (84.4%) and amikacin (80.0%). Similarly, collared peccaries (*Pecari tajacu*), oncillas (*Leopardus tigrinus*), taurus (*E. barbara*), white-faced monkeys (*C. albifrons*), tufted capuchins (*Cebus apella*), howler monkeys (*Alouatta seniculus*), tapirs (*T. terrestris*) and grey tits (*Saguinus leucopus*) serve as reservoirs for *E. coli* strains resistant to ampicillin and ampicillin/sulbactam, oxytetracycline (OT), ciprofloxacin, cefotaxime and ceftazidime (Medina et al., 2017; Vargas et al., 2010).

In Mexico, Canada and Belgium, similar studies have been conducted on captive mammals. Research in Mexico revealed that spider monkeys (*Ateles geoffroyi*), Baird's tapirs (*Tapirus bairdii*), jaguars (*P. onca*), cougars (*Puma concolor*) and ocelots (*L. pardalis*) showed *E. coli* isolates resistant to antibiotics such as ampicillin, amoxicillin/clavulanic acid, chloramphenicol, sulfadiazine, ciprofloxacin and tetracycline (Cristóbal-Azkarate et al., 2014). In Canada, a study on raccoons (*Procyon lotor*) from the Toronto Zoo found these animals capable of harbouring *E. coli* with reduced susceptibility to tetracycline, amoxicillin/clavulanic acid, cefoxitin, ceftiofur, ceftriaxone and ciprofloxacin, whereas research in Belgium conducted on jaguars (*P. onca*), Bengal tigers (*P. tigris tigris*) and spectacled bears (*Tremarctos ornatus*) revealed *E. coli* with a 64% multidrug resistance rate, showing resistance to trimethoprim/sulfamethoxazole, tetracycline, doxycycline, streptomycin, trimethoprim, enrofloxacin and gentamicin (De Witte et al., 2021; Jardine et al., 2012). This study findings mirror those of other research on the same or related species; however, our results also indicate a lower percentage of multidrug resistance compared to other studies.

These differences are marked by various factors. A study focused on the factors associated with AMR of *E. coli* isolates in captive mammals, finding that animals previously treated with antibiotics showed higher resistance rates than untreated animals. In their research, isolates exhibited resistance to ampicillin, aminoglycoside antibiotics, tetracycline and trimethoprim, despite these antibiotics being rarely administered (Ishihara et al., 2012).

In the Eco Zoo 'San Martín' in Baños de Agua Santa, a similar situation occurs. Antibiotics that showed resistance included ampicillin, ceftriaxone, cefuroxime, trimethoprim/sulfamethoxazole, ceftazidime and ciprofloxacin. However, the first-line antibiotics used for veterinary treatment for this zoo are enrofloxacin, doxycycline, ceftriaxone, ampicillin/sulbactam, metronidazole, ampicillin and tobramycin. This phenomenon had been previously observed in studies focused on dairy-producing animals (Azabo et al., 2022), in samples from humans and food animals (Tadesse et al., 2012) and even in bacteria isolated from deceased animals (Bourély et al., 2019).

A research conducted in a wildlife rescue centre, where they observed various antibiotic resistance profiles in *E. coli*, the bacteria demonstrated resistance to more than nine antibiotics from different families (Jorquera et al., 2021). Antibiotic treatment duration was mentioned as a factor in multiresistance, with the period of treatment influencing the number of antibiotics to which the strains acquired resistance. It is important to note that the results demonstrate that the highest resistance rates are observed in those antibiotics that have been used for a longer time in human and veterinary medicine (Andersson et al., 2021; Caneschi et al., 2023).

The antibiotic resistance patterns observed in our study may be influenced by several key factors. First, differences in dietary practices play a significant role, as food can act as a carrier of antibiotic-resistant bacteria, making it crucial to analyse the potential impact of diet on resistance patterns (Barragán-Fonseca et al., 2022; Tubón et al., 2022). Second, the diverse origins of the zoo animals, which come from various places with different veterinary practices and histories of human and animal contact, make it challenging to trace the sources of resistant bacteria they may carry (Furlan et al., 2024). Third, vectors such as flies and birds, as well as zoo personnel, could contribute to the transmission of antibiotic-resistant bacteria within the zoo (Yin et al., 2022). Additionally, although the use of antibiotics as growth promoters in Ecuador is common and has led to the presence of multidrug-resistant bacteria in food (Braykov et al., 2016; Ortega-Paredes et al., 2020); the closed environment of zoos presents a unique opportunity to study the influence of these factors on the evolution and dissemination of resistant bacteria in a more controlled setting.

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TABLE 1 Phenotypic and genotypic characteristics of *Escherichia coli* strains that harbour extended-spectrum beta-lactamase (ESBL) genes from captive mammals.

Isolate	Origin	ESBL positive	Non-beta-lactam antibiotic resistance phenotype	ESBL genes	Conjugation to *E. coli* J53
E4	*Panthera tigris tigris*	–	–	bla_{TEM}	–
E5	*P. tigris tigris*	–	CN-SXT	bla_{TEM}, $bla_{CTX\text{-}M\ Group\ 1}$	–
E6	*P. tigris tigris*	–	–	bla_{TEM}	–
E7	*P. tigris tigris*	+	F-SXT	bla_{TEM}, $bla_{CTX\text{-}M\ Group\ 9}$	–
E24	*Potos flavus*	–	CIP	bla_{TEM}	–
E32	*Eira barbara*	+	SXT	$bla_{CTX\text{-}M\text{-}Group\ 1}$	–
E33	*Procyon cancrivorus*	+	FF	$bla_{CTX\text{-}M\text{-}Group\ 1}$	–
E35	*Cebus albifrons*	–	SXT	$bla_{CTX\text{-}M\text{-}Group\ 9}$	+
E36	*C. albifrons*	+	SXT	$bla_{CTX\text{-}M\text{-}Group\ 9}$	+
E37	*C. albifrons*	+	–	$bla_{CTX\text{-}M\text{-}Group\ 9}$	+
E38	*C. albifrons*	+	–	$bla_{CTX\text{-}M\text{-}Group\ 9}$	+
E39	*C. albifrons*	+	FF	$bla_{CTX\text{-}M\text{-}Group\ 1}$	+
E46	*Dicotyles tajacu*	+	–	$bla_{CTX\text{-}M\text{-}Group\ 9}$	+
E62	*Leopardus pardalis*	+	–	$bla_{CTX\text{-}M\text{-}Group\ 1}$	+
E63	*L. pardalis*	+	SXT	$bla_{CTX\text{-}M\text{-}Group\ 1}$	+
E65	*L. pardalis*	+	SXT-CIP	$bla_{CTX\text{-}M\text{-}Group\ 1}$	+
E66	*L. pardalis*	–	CIP	$bla_{CTX\text{-}M\text{-}Group\ 1}$	–
E67	*L. pardalis*	+	SXT-CIP	$bla_{CTX\text{-}M\text{-}Group\ 1}$	+
E76	*L. pardalis*	+	SXT-CIP	$bla_{CTX\text{-}M\text{-}Group\ 1}$	+
E77	*L. pardalis*	+	SXT	$bla_{CTX\text{-}M\text{-}Group\ 1}$	+
E78	*L. pardalis*	–	SXT-CIP	$bla_{CTX\text{-}M\text{-}Group\ 1}$	–
E79	*L. pardalis*	+	FF-CIP	$bla_{CTX\text{-}M\text{-}Group\ 1}$	+
E80	*L. pardalis*	+	SXT-CIP	$bla_{CTX\text{-}M\text{-}Group\ 1}$	+
E68	*Cebus capucinus/Callicebus discolor*	+	SXT	$bla_{CTX\text{-}M\text{-}Group\ 9}$	–
E69	*C. capucinus/C. discolor*	–	SXT-CIP	$bla_{CTX\text{-}M\text{-}Group\ 9}$	–
E72	*Panthera onca*	–	SXT	bla_{TEM}	–
E73	*P. onca*	–	–	bla_{TEM}	–
E75	*P. onca*	–	CN-SXT-CIP	$bla_{CTX\text{-}M\text{-}Group\ 1}$	–
E98	*Tapirus terrestris*	+	–	$bla_{CTX\text{-}M\text{-}Group\ 1}$	–
E99	*T. terrestris*	–	–	$bla_{CTX\text{-}M\text{-}Group\ 9}$	–

Abbreviations: CIP, ciprofloxacin; CN, gentamicin; F, nitrofurantoin; FF, Fosfomycin; SXT, trimethoprim/sulfamethoxazole.

4 | CONCLUSIONS

The investigation revealed that 30 out of 90 *E. coli* isolates (33.33%) exhibit susceptibility or intermediate susceptibility to the antibiotics used, whereas 60 isolates (66.67%) show resistance to at least one antibiotic. Notably, a frequent detection of organisms resistant to carbapenemic and β-lactam antibiotics was observed, indicating a potential public health issue. Moreover, 35.56% of isolates exhibited resistance to more than three families of antibiotics. This finding underscores the role of captive mammals in zoos as indicators of AMR in the environment and highlights the need for focused AMR studies in Ecuador to control and prevent the progression of this phenomenon. These findings also emphasize the importance of reviewing hygienic and sanitization practices in zoo habitats to reduce associated zoonotic risks.

The interplay of various factors, such as animal diets, intrinsic characteristics and antibiotic treatments, greatly influences the occurrence of antibiotic resistance. Animals with carnivorous diets and prior antibiotic treatments showed more notable resistance profiles. This complex interplay underscores the need for an integrated approach combining veterinary care, habitat management and public awareness. Such an approach, focusing on responsible antibiotic use, hygienic conditions and continuous monitoring, is essential for Eco Zoo San Martin to safeguard the health of its captive wildlife and prevent these animals from becoming reservoirs of antibiotic-resistant bacteria.

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Furthermore, these findings signal the need to implement better practices for handling faeces and spilled food. By enhancing waste management protocols, the zoo can minimize the risk of contamination and the spread of antibiotic-resistant bacteria. Effective waste collection and disposal are crucial to reducing the persistence of these bacteria in the environment. These improved waste management strategies are integral to maintaining a safe and healthy environment for both the animals and the zoo visitors, thereby safeguarding public health.

AUTHOR CONTRIBUTIONS

Anabell Medina, Yadira Vega and William Calero-Cáceres designed the study. Anabell Medina, Jennifer Medina, Rosa N. López, Patricio Vayas, Joyce Soria, Christian Velásquez and Lissette Sánchez-Gavilanes conducted the preliminary data analyses, selective isolation, bacterial identification and antimicrobial susceptibility testing. Anabell Medina, Jennifer Medina, Rosa N. López and Lissette Sánchez-Gavilanes performed the detection of antibiotic resistance genes, GTG-fingerprinting and transconjugation assays. Carlos Bastidas-Caldes conducted the Sanger sequencing and assisted with data analysis. William Calero-Cáceres and Yadira Vega supervised the entire project. William Calero-Cáceres and Anabell Medina wrote the manuscript, with all authors contributing to its revision and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no known conflicts of interest or personal relationships that could have influenced the work reported in this paper.

DATA AVAILABILITY STATEMENT

Specific primers and conditions for each gene are detailed in Table S1. Table S2 provides information on the origin, habitat, common name and scientific name of the animals from which the isolates were obtained. All other data are presented within the document.

ETHICS STATEMENT

This study was conducted with approval from the Ministry of Environment of Ecuador (contract MAATE-DBI-CM-2023-0278). The research did not directly involve animals but utilized their faeces, with zoo specialists overseeing the sampling process to ensure minimal disturbance to the animals.

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